

SUMMARY TABLE: BIOLOGY GRADE 11

Sub-Strand 1.3: Taxonomy II — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 11	
Sub-Strand	Sub-Strand 1.3: Taxonomy II
Driving Question	DRIVING QUESTION: How can we identify and classify living organisms accurately?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Why classify organisms?	We observed 13 local organisms (maize, bean plant, grass, fern, moss, mushroom, mosquito, butterfly, earthworm, snail, chicken, lizard, goat) and noted that they differ in structure, size, body covering, movement and reproduction. Different groups sorted them in different ways (by size, colour, habitat, usefulness).	Organisms can be grouped using similarities and differences, but groupings made with different rules do not agree. Scientists need a standard system of classification based on shared, observable characteristics.	Because everyone chose different rules, no single grouping worked for everyone. A standard system uses the same observable characteristics so that any person anywhere can identify and study an organism and get the same result. We still need to find out how scientists decide which characteristics to use.
Lesson 2 Kingdom Plantae — Bryophyta	Describe what you observed in the moss with a hand lens: its size, the base (rhizoids), the stem-like and leaf-like parts, any spore capsules, where it was growing, and how it differed from maize or bean plants.	State the characteristics that identify Division Bryophyta (e.g. small size, rhizoids instead of true roots, no vascular tissue, spores in capsules, damp habitats) and how they differ from other plants.	Explain why moss stays small and grows in damp places, and how this evidence helps scientists classify plants more accurately than grouping by size, colour or use. What new question do you now have about how other plants differ from bryophytes?
Lesson 3 Pteridophyta and Spermatophyta	We observed that ferns have true roots, a rhizome and fronds with brown sori on the underside, while maize and bean plants have roots, stems, leaves and flowers that produce seeds (grains and pods). Moss has no true roots and no vascular tissue.	Pteridophytes reproduce by spores and have vascular tissue but no seeds or flowers; spermatophytes reproduce by seeds and have vascular tissue; bryophytes lack vascular tissue and true roots. These are shared, observable features anyone can check.	These features help explain our phenomenon because they give everyone the same rules for grouping plants, instead of personal groupings by size, colour or use as food. We still need a step-by-step way to identify an unknown plant.
Lesson 4 Using Dichotomous Keys	We used a sample dichotomous key and a ruler to identify local plants (maize, bean, grass, fern, moss, sukuma wiki) and found	A dichotomous key uses pairs of contrasting, observable characteristics, in numbered couplets, so that each step has	A key gives the same answer to everyone because it uses clear, observable features rather than personal opinions such as size,

	that different group members reached the same name when they followed the same couplets.	only two choices and leads to one identification. Features such as true roots, stems and leaves, fronds, seeds, and parallel versus net veins help place plants into Bryophyta, Pteridophyta and Spermatophyta (monocots and dicots).	colour or usefulness. This is how scientists can organise the organisms around our school so that anyone can identify them. New question: can the same method be used to identify animals?
Lesson 5 Kingdom Animalia — major phyla	We observed that the earthworm has a soft segmented body with no legs, the snail has a soft body with a shell, the mosquito and butterfly have jointed legs and a hard outer covering (exoskeleton), and the chicken, lizard and goat all have a backbone but different body coverings (feathers, scales, hair).	We learned that animals are classified into major phyla (Annelida, Mollusca, Arthropoda, Chordata) using observable body features such as segmentation, shell, jointed legs with an exoskeleton, and a backbone, not by size, colour, habitat or usefulness.	We can explain that the school animals can be organised so anyone can identify them because phylum placement is based on shared, checkable body features. For example, the goat and the chicken are both Chordata because they both have a backbone, while the mosquito is Arthropoda because it has jointed legs and an exoskeleton. We still wonder how animals within the same phylum, such as the chicken, lizard and goat, are further classified.
Lesson 6 Classes of Kingdom Animalia	We observed that a goat, chicken, lizard and fish (all Chordata) differ in body covering (hair, feathers, scales), limbs, respiration (lungs, gills) and reproduction (eggs or live young), and that a mosquito, butterfly and crab (all Arthropoda) differ in number of legs, body parts and antennae.	We learned that phyla can be divided further into classes using more specific characteristics: Chordata into Pisces, Amphibia, Reptilia, Aves and Mammalia; Arthropoda into Insecta, Arachnida, Crustacea and Myriapoda.	We can explain that the school animals can be organised into classes using shared, observable features (not size, colour or habitat), so anyone can classify them the same way, but we still need a reliable method to identify an unknown animal accurately, which leads to the next lesson.
Lesson 7 Dichotomous keys for animals and viruses	What did we observe? (Record how your dichotomous key worked when another group used it, which couplets were clear or confusing, which animals fell in Chordata and Arthropoda, and what the virus diagrams showed about structure.)	What did we learn? (A dichotomous key uses paired contrasting observable features to identify organisms reliably; viruses are non-cellular, have a protein coat and genetic material, and reproduce only inside host cells; viruses are important in health, agriculture and biotechnology.)	How does this help explain the phenomenon? (Explain why a key gives the same answer for everyone while groupings by size, colour or usefulness do not, and how knowing about viruses shows that classification must be based on shared observable characteristics. Raise a question for Lesson 8 on biodiversity.)
Lesson 8 Synthesis and explanation	We observed that organisms from our school environment (maize, bean, fern, moss, mushroom, mosquito, butterfly, earthworm, snail, chicken, lizard, goat) differ in structure, body covering, movement and reproduction, and that groups using size, colour or usefulness produced different groupings. When we used shared, observable characteristics and dichotomous keys, different groups reached the same results.	We learned that plants are classified into Bryophyta, Pteridophyta and Spermatophyta (Gymnospermae and Angiospermae; monocots and dicots), and animals into phyla such as Chordata and Arthropoda and then classes. We learned how to build and use dichotomous keys and that viruses are non-cellular but important in health, agriculture and biotechnology.	We can explain that organisms can be identified and classified accurately by using a standard system based on shared, observable characteristics and tools such as dichotomous keys, so that anyone anywhere can study and communicate about biodiversity. This replaces our Lesson 1 groupings, which varied from group to group.

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